

Problem Formulation & Environment

Classification :-

1. Problem Formulation :

- Initial State :- The student submits the MCQ answer sheet. All questions are unchecked and Total marks = 0.
- Actions :- Read student's answer, compare with answer key, mark correct or wrong, add marks, move to next question, display final result.
- Transition Model :- When an answer is checked, the question status changes from unchecked to checked. If the answer is correct, marks are added. If incorrect, no marks are added.
- Path Cost :- Total Time taken to check the paper. The goal is to finish checking in the shortest time with 100% accuracy.
- Goal State :- All questions on the paper are checked, Total marks are counted and the final result is shown.

→ State Space :- All possible states from start to end (e.g: question 1 checked, question 2 checked, mark obtained at each step).

→ Search Space :- The step-by-step sequence the agent follows. Starting from question 1, checking each answer one by one, until the final score is reached.

→ Agents :- Single agent. Only the software / system is checking and grading the paper.

2. Environmental Properties :

→ Observability : Fully Observable

The agent can clearly see the full answer sheet and answer key, nothing is hidden.

→ Determinism : Deterministic

Every action has a fixed result. If the answer matches the key, it is correct. If not, it is wrong. There is no guessing.

→ Episodes : Episodic

Checking one student's paper does not depend on checking the previous or next student's paper.

→ Change : Static

The paper and the answer key remains the same, while the agent is checking.

→ Values : Discrete

The question, marks and options are separate and easy to count.

→ Agents : Single - Agent

Only one agent is doing the work of checking the exam paper.

3. State Type :

Single State - Most Suitable

The agent always knows its current state. The environment is fully visible, unchanging and deterministic, meaning every single check gives a clear and predictable result without any confusion.